

PRASHANT SHAH

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EDUCATION

Ramapo College of New Jersey

B.S. Data Science & Computer Science, Minor in Mathematics | GPA: 3.8/4.0

Recognitions: Best Research Award, ICPC NY Honorable Mention, Presidential Scholarship

Mahwah, NJ

May 2026

EXPERIENCE

Software Engineering Intern

May 2025 - Aug 2025

Trimble Inc.

Princeton, NJ

- Integrated service-configuration data across TMW Suite, environment databases, internal APIs, and authentication boundaries so distributed configuration streams became visible and editable in one workflow.
- Built C# .NET components, REST APIs, SQL validation queries, and Azure authentication tooling to help engineers catch multi-environment conflicts before downstream support issues.
- Worked inside existing enterprise service boundaries, translating scattered configuration behavior into workflow logic without breaking downstream transportation-platform services.

Data Engineer (Contract)

Sep 2025 - Present

Center for Food Action

Mahwah, NJ

- Built a Flask/PostgreSQL operations platform integrating barcode scans, UPC/price lookup services, donor/storage records, and PantrySoft handoffs into validated intake workflows. Featured in Ramapo News and News 12 NJ.
- Implemented SQLAlchemy models, review-before-save states, blocked-export logic, reconciliation queries, and PantrySoft receive/export contracts so non-technical users could catch bad mappings or quantities before data left the system.
- Added Playwright/API validation paths and reporting-ready records for pricing, inventory movement, service trends, and client-origin analytics, connecting backend reliability to operational decisions.

Research Assistant - Retrieval Systems (Link)

May 2025 - May 2026

Ramapo College of New Jersey

Mahwah, NJ

- Researched content-intent retrieval where topical semantics and affective intent conflict; built Krag on MovieLens 25M with User, Movie, and Emotion nodes, weighted EVOKES edges, and graph/text ranking.
- Used a Graph Transformer aligned to Sentence-BERT plus ablations to show emotion edges were about 2.3x more necessary under dissonance, turning retrieval failures into system-level diagnosis.

Software Engineering Intern, Business Analytics

Jun 2024 - Aug 2024

Novartis Pharmaceuticals Corporation

East Hanover, NJ

- Built PostgreSQL data models and Azure Functions workflows to standardize project, budget, approval-status, and team-allocation data across 100+ internal initiatives in a regulated pharmaceutical environment.
- Delivered Power BI/Tableau views and data-cleanup workflows that reduced a recurring 2-day reporting workload into refreshable operational views for clinical operations stakeholders.

PROJECTS

Hermes Tooling and Plugin Integrations (Link)

- Built a Python/SQLite web cockpit where the dashboard owns jobs, evidence, contacts, materials, statuses, and follow-ups while Hermes runs as the API/tool engine behind streaming sessions and project-plugin context.
- Integrated Hermes API sessions, SSE streaming, native tool dispatch, ATS hydration, contact caching, Gmail draft creation, Typst/TeX material compilation, and provenance records so human actions and agent runs work from the same structured state.

Atlas LMS - AI-native learning platform (Link)

- Won HackRamapo 2026 with a Vercel AI SDK course-management agent whose typed tools run parameterized SQL, never model-generated SQL, over a Canvas-modeled PostgreSQL schema with live ETL for courses, assignments, and deadlines.
- Architecting Atlas as a multi-tenant Next.js/PostgreSQL platform with Drizzle, Auth.js, pgvector memory, generative UI, org-scoped data, and human approval gates for AI-authored course changes.

SKILLS

Languages: Python, C#, TypeScript, JavaScript, Java, C/C++, SQL (PostgreSQL, MySQL, SQLite), Bash

Backend & APIs: .NET, REST APIs, Flask, FastAPI, Node.js, SQLAlchemy, integrations, auth/RBAC, API contracts

Cloud/Data: Azure, Azure Functions, AWS, Vertex AI, Docker, Linux, GitHub Actions, PostgreSQL, pgvector, ETL

AI/Product: LLM APIs, tool calling, agents, RAG, embeddings, generative UI, React, Next.js, Vercel AI SDK

Practices: Testing, Playwright, Selenium, pytest, CI/CD, code review, schema design, validation, Agile/Scrum